

Answers to the ques: 1

$$X \rightarrow a$$

$$X \rightarrow Y$$

$$Z \rightarrow d$$

$$Z \rightarrow XYZ$$

$$Y \rightarrow c$$

$$Y \rightarrow \epsilon$$

$$X \rightarrow \overset{1}{a} | \overset{2}{Y}$$

$$Y \rightarrow \overset{3}{c} | \overset{4}{\epsilon}$$

$$Z \rightarrow \overset{5}{d} | \overset{6}{XYZ}$$

variable	First	Follow
X	{a, c, ϵ }	{c, \$}
Y	{c, ϵ }	{d, a, c, \$}
Z	{d, a, c, ϵ }	{d, \$}

Parse Table:

	a	c	d	\$
X	1/2	2	2	2
Y	4	3/4	4	4
Z	6	6	5	6

Answer to the ques. no: 02

$$S \rightarrow A^1 a A^2 b | B b$$

$$A \xrightarrow{3} \epsilon, B \xrightarrow{4} \epsilon$$

$$\text{First}(A) = \{\epsilon\}$$

$$\text{First}(B) = \{\epsilon\}$$

$$\text{First}(S) = \{a\}$$

$$\left. \begin{array}{l} \text{Follow}(S) = \{\$ \} \\ \text{Follow}(A) = \{a\} \\ \text{Follow}(B) = \{b\} \end{array} \right\}$$

Parse table:

	a	b	\$
A	3		
B		4	
S	1	2	

Ans. to the ques: 03

$$C \xrightarrow{1} PF \text{ class id } XY$$

$$P \xrightarrow{2} \text{public} \mid \epsilon^3$$

$$F \xrightarrow{4} \text{final} \mid \epsilon^5$$

$$X \xrightarrow{6} \text{extends} \mid \epsilon^7$$

$$Y \xrightarrow{8} \text{implements} \mid \epsilon^9$$

$$I \xrightarrow{10} \text{id } J$$

$$J \xrightarrow{11} I \mid \epsilon^{12}$$

$$\text{First}(C) \rightarrow \{ \text{public}, \text{final}, \text{class}, \epsilon \}$$

$$\text{First}(P) \rightarrow \{ \text{public}, \epsilon \}$$

$$\text{First}(F) \rightarrow \{ \text{final}, \epsilon \}$$

$$\text{First}(X) \rightarrow \{ \text{extends}, \epsilon \}$$

$$\text{First}(Y) \rightarrow \{ \text{implements}, \epsilon \}$$

$$\text{First}(I) \rightarrow \{ \text{id} \}$$

$$\text{First}(J) = \{ \text{id}, \epsilon \}$$

compiler design {sec: E}

Follow (c) = { \$ }

Follow (P) = { final, class }

Follow (F) = { class }

Follow (X) = { extends }

Follow (Y) = { implements }

Follow (I) = { \$ }

Follow (J) = { \$ }

Parse table

	class	id	public	final	extend	impleme nt	\$
c			1				
P	3		2	3			
F	5			4			
X					6/7		
Y						8/9	
I		10					
J		11					12

$$S \xrightarrow{1} \bar{x} a$$

$$x \xrightarrow{2} \bar{y} z$$

$$y \xrightarrow{3} \bar{b} \epsilon$$

$$z \xrightarrow{5} \bar{c} \epsilon$$

$$\text{first}(S) \rightarrow \{\epsilon\}$$

$$\text{first}(x) \rightarrow \{\epsilon\}$$

$$\text{first}(y) \rightarrow \{\epsilon\}$$

$$\text{first}(z) \rightarrow \{c, \epsilon\}$$

$$\text{follow}(S) \rightarrow \{\$\}$$

$$\text{follow}(x) \rightarrow \{a\}$$

$$\text{follow}(y) \rightarrow \{c\}$$

$$\text{follow}(z) \rightarrow \{\$\}$$

Parse table

	a	c	\$
S	1		
x		2	
y		4	3
z		5	6

The end.